

$$F[\phi] = \int d^D x \int d^D y \phi(x) K(x, y) (\phi(y))^2 e^{\omega}$$

$$\omega = i \int d^D x \dot{\phi}_x \phi_x + \int d^D x d^D y d^D z d^D \omega A(x, y, z, \omega) (\partial \phi(x))^2 \phi_y \phi_z \phi_\omega$$

(i) Find $\frac{\delta F}{\delta \phi(x)}$

$$\delta F = (\delta \phi_x K_{xy} \phi_y^2) e + 2 \phi_x K_{xy} \phi_y \delta \phi_y e + \phi_x K_{xy} \phi_y^2 e^{\omega} \delta \omega$$

$$\delta \omega = i \dot{\phi}_x \delta \phi_x + 2 \partial \phi_x \delta (\partial \phi_x) A_{xyz\omega} \phi_y \phi_z \phi_\omega + 3 (\partial \phi_y)^2 A_{yxz\omega} \phi_\omega \phi_z \delta \phi_x$$

$$\delta \omega = (i \dot{\phi}_x + - 2 \partial_x^m (\partial_x^m \phi_x A_{xyz\omega}) \phi_y \phi_z \phi_\omega + 3 (\partial \phi_y)^2 A_{yxz\omega} \phi_\omega \phi_z) \delta \phi_x$$

∴

$$\delta F = \int d^D x [K_{xy} \phi_y^2 + 2 \phi_y K_{yx} \phi_x$$

$$\phi_x K_{\alpha\beta} \phi_\beta (i \dot{\phi}_x - 2 \partial_x^m (\partial_x^m \phi_x A_{xyz\omega}) \phi_y \phi_z \phi_\omega + 3 (\partial \phi_y)^2 A_{yxz\omega} \phi_\omega \phi_z) \delta \phi_x e^{\omega}$$

$$\frac{\delta F}{\delta \phi} = ()$$

$$\phi(x) = \int \frac{d^3 k}{(2\pi)^3} \frac{1}{2\omega_k} (e^{i k x} \hat{a}_k + e^{-i k x} \hat{a}_k^\dagger)$$

$$\langle \phi | \hat{a}_k | \psi \rangle = \int d^3 x e^{-i k x} \left(\sqrt{-v^2 + m^2} \phi + \frac{\delta}{\delta \phi(x)} \langle \psi | \phi \rangle \right)$$

$$= \left(\sqrt{k^2 + m^2} \hat{\phi}(k) + \frac{\delta}{\delta \hat{\phi}(-k)} \right) \langle \phi | \psi \rangle$$

$$|\phi\rangle = A e^{-\frac{1}{2} \int d^3 x \phi(x) \sqrt{-v^2 + m^2} \phi(x)}$$

$$\phi(x) = \int \frac{d^3k}{(2\pi)^3} e^{ikx} \tilde{\phi}(k)$$

$$\begin{aligned} \langle \phi | 0 \rangle &= A e^{-\frac{1}{2} \int d^3x \int \frac{d^3k}{(2\pi)^3} \int \frac{d^3p}{(2\pi)^3} e^{ix(p+k)} \sqrt{p^2+m^2} \tilde{\phi}(k) \tilde{\phi}(p)} \\ &= A e^{-\frac{1}{2} \int \frac{d^3k}{(2\pi)^3} \frac{\int d^3p}{(2\pi)^3} \delta(p+k) \sqrt{p^2+m^2} \tilde{\phi}(k) \tilde{\phi}(p)} \\ &= A e^{-\frac{1}{2} \int \frac{d^3k}{(2\pi)^3} \tilde{\phi}(k) \sqrt{k^2+m^2} \tilde{\phi}(-k)} \end{aligned}$$

$$Q: \langle \phi | a^\dagger | 0 \rangle = \left(\sqrt{k^2+m^2} \tilde{\phi}(k) + \frac{\delta}{\delta \tilde{\phi}(-k)} \right) \langle \phi | 0 \rangle$$

$$\frac{\delta}{\delta \tilde{\phi}(-k)} \langle \phi | 0 \rangle = \tilde{\phi}(k) \sqrt{k^2+m^2} \langle \phi | 0 \rangle$$

$$\langle \phi | a^\dagger | 0 \rangle = 2 \tilde{\phi}(k) \sqrt{k^2+m^2} \langle \phi | 0 \rangle$$

$$\langle \phi | a^\dagger | 0 \rangle = 2A \tilde{\phi}(k) \sqrt{k^2+m^2} e^{-\frac{1}{2} \int \frac{d^3k}{(2\pi)^3} \tilde{\phi}(k) \omega_k \tilde{\phi}(-k)}$$

$$Q: \langle \phi | a_{k_2}^\dagger a_{k_1}^\dagger | 0 \rangle?$$

$$\langle \phi | a_{k_2}^\dagger \int D\phi' |\phi'\rangle \langle \phi' | a_{k_1}^\dagger | 0 \rangle$$

$$= \int D\phi' \langle \phi | a_{k_2}^\dagger | \phi' \rangle \langle \phi' | a_{k_1}^\dagger | 0 \rangle$$

$$\int D\phi' \left(\int d^3k e^{ikx_{k_1}} \sqrt{-\nabla^2+m^2} \phi_1 + \frac{\delta}{\delta \phi_{k_1}} \right) (\langle \phi | \phi' \rangle) \langle \phi' | 0 \rangle$$

$$= \int D\phi' \left(\int d^3k e^{ikx_{k_1}} \sqrt{-\nabla^2+m^2} \phi_1' + \frac{\delta}{\delta \phi_{k_1}'} \right) \delta(\phi - \phi') \langle \phi' | 0 \rangle$$

$$\frac{\delta}{\delta \phi_{k_1}'} \delta(\phi - \phi') = \int D\pi e^{i \int d^3y \pi(y) (\phi_y - \phi_y')}$$

$$= \int D\pi e^{i \int d^3y \pi(y) (\phi_y - \phi_y')}$$

$$= i \int D\pi(y) \pi(y) e^{i \int d^3y \pi(y) (\phi_y - \phi_y')}$$

$$\neq \int D\phi' \int D\pi(y) \pi(x) e^{i\pi_y \phi_y} e^{-i\pi_y \phi'_y}$$

$$= \int D\pi(y) \pi(x) e^{i\pi_y \phi_y} \int D\phi'_x e^{-i\pi_y \phi'_y}$$

$$= \int D\pi_y \pi(x) e^{i\pi_y \phi_y} \delta(\pi_y - 0)$$

$$= 0$$

• δ - too complicated:

$$\langle \phi | \hat{a}_{k_1}^\dagger \hat{a}_{k_2}^\dagger | 0 \rangle = \left(\int d^3x e^{i x k_1} \sqrt{-\nabla^2 + m^2} \phi(x) + \frac{\delta}{\delta \phi(x)} \right) \left(2 \int d^3y e^{i y k_2} \sqrt{-\nabla^2 + m^2} \phi(y) \right) \langle \phi | 0 \rangle$$

$$= -2 \int d^3x e^{i k_1 + k_2 x} \sqrt{m^2 + k^2} \langle \phi | 0 \rangle + 4(\dots) \langle \phi | 0 \rangle$$

$$P(q, p) = Z^{-1} e^{-\beta H(q, p)}$$

$$Z = -kT \ln(Z)$$

$$\int d^D q \int \frac{d^D p}{(2\pi\hbar)^D} P(q, p) = 1$$

$$Z = \int d^D q \int \frac{d^D p}{(2\pi\hbar)^D} e^{-\beta H(q, p)}$$

$$H = \frac{1}{2m} p^2 + \frac{1}{2} m \omega^2 \vec{x}^2$$

$$Z = \int d^D q \int \frac{d^D p}{(2\pi\hbar)^D} e^{-\beta p^2 / 2m} e^{-\beta \frac{1}{2} m \omega^2 \vec{x}^2}$$

$$D=3$$

$$Z = \int d^3 q \int \frac{d^3 p}{(2\pi\hbar)^3} e^{-\beta p^2 / 2m} e^{-\beta \frac{1}{2} m \omega^2 q^2}$$

$$Z = \left(\int d^3 q e^{-\beta \frac{1}{2} m \omega^2 q^2} \right) \left(\int \frac{d^3 p}{(2\pi\hbar)^3} e^{-\beta / 2m p^2} \right)$$

$$\left(\int d^3 q e^{-\beta \frac{1}{2} m \omega^2 q^2} \right)^{3/2} \left(\int d^3 p e^{-\beta / 2m p^2} \right)^{3/2}$$

$$\begin{aligned}
 &= \left(\frac{1}{(\beta/2)m\omega} \right) \left(\frac{1}{(2\pi\hbar)^3} \right) \left(\frac{\pi^2 \omega^2}{\beta} \right) \\
 &= \frac{1}{(2\pi\hbar)^3} \left(\frac{\pi^2 \omega^2}{\beta^2 m \omega^2} \right)^{3/2} \\
 &= \frac{1}{(2\pi\hbar)^3} \left(\left(\frac{\pi^2}{\beta \omega} \right)^3 \right) = \frac{1}{(\hbar\beta\omega)^3}
 \end{aligned}$$

$$Z = (\hbar\beta\omega)^{-3} = \left(\frac{\hbar\omega}{kT} \right)^3$$

$$F = -kT \ln(Z)$$

$$F = -3kT \ln \left(\frac{\hbar\omega}{kT} \right) = 3\beta^{-1} \ln(\hbar\omega\beta)$$

(b) Let $\hat{A}: \mathcal{Y}^2(\mathcal{H}) \rightarrow \mathcal{Y}^2(\mathcal{H})$

then

$$\text{Tr}(\hat{A}) = \sum_n \langle \phi_n | \hat{A} | \phi_n \rangle$$

$$\sum_n |\phi_n\rangle \langle \phi_n| \cong \mathbb{I}$$

* Energy eigen: $|\phi_n\rangle =: |n\rangle$

$$Z(\beta) = \text{Tr}(e^{-\beta\hat{H}}) = \sum_n \langle n | e^{-\beta\hat{H}} | n \rangle$$

$$\hat{\mathbb{I}} = \int d^D q |q\rangle \langle q|$$

$$= \int d^D q d^D q' \sum_n \langle n | q \rangle \langle q | e^{-\beta\hat{H}} | q' \rangle \langle q' | n \rangle$$

$$= \int d^D q d^D q' \langle q | \sum_n |n\rangle \langle n| | q' \rangle \langle q | e^{-\beta\hat{H}} | q' \rangle$$

$$= \int d^D q d^D q' \delta(q - q') \langle q | e^{-\beta\hat{H}} | q' \rangle$$

$$Z(\beta) = \int d^D x \langle x | e^{-\beta\hat{H}} | x \rangle$$

$$\langle x_f | e^{-i\hat{H}T/\hbar} | x_i \rangle = \int_{x(0)=x_i}^{x(T)=x_f} \mathcal{D}x(t) \int \mathcal{D}p(t) e^{\frac{i}{\hbar} \int_0^T dt \left(\vec{p} \frac{d\vec{x}}{dt} - \vec{H}(\vec{x}, \vec{p}) \right)}$$

$$i) \text{ let } x_i = x_f = x$$

$$\langle x | e^{-i\hat{H}T/\hbar} | x \rangle$$

$$ii) iT/\hbar = \beta$$

or

$$iT = \hbar \beta \equiv \mathcal{Z}$$

$$\langle x | e^{-\beta \hat{H}} | x \rangle = \int_{x(0)=x_i}^{x(\hbar\beta)=x_f} \mathcal{D}x(\tau) \int \mathcal{D}p(\tau) e$$

$$t = -i\tau$$

$$dt = -i d\tau$$

$$\Rightarrow e^{\frac{i}{\hbar}(-i) \int_0^{\hbar\beta} d\tau \left(\vec{p} \frac{d\vec{x}}{(-i)d\tau} - \hat{H}(x, p) \right)}$$

$$= e^{\frac{1}{\hbar} \int_0^{\hbar\beta} d\tau \left(i \vec{p} \frac{d\vec{x}}{d\tau} - \hat{H}(x, p) \right)}$$

$$\therefore \mathcal{Z}(\beta) = \int d^3x_i \langle x_i | e^{-\beta \hat{H}} | x_i \rangle$$

$$= \int d^3x_i \int_{x(0)=x_i}^{x(\hbar\beta)=x_i} \mathcal{D}x_i(\tau) \int \mathcal{D}p(\tau) e^{\frac{1}{\hbar} \int_0^{\hbar\beta} d\tau \left(i \vec{p} \cdot \frac{d\vec{x}}{d\tau} - H(q, p) \right)}$$

$$= \int d^3x_i \lim_{N \rightarrow \infty} \left(\frac{1}{T!} \int_{-\infty}^{\infty} d^3x_i \right) \lim_{N \rightarrow \infty} \left(\frac{1}{T!} \int_{-\infty}^{\infty} d^3p_j (2\pi\hbar)^3 \right)$$

$$\times \exp \left[\frac{1}{\hbar} \Delta\tau \sum_j \left(\vec{p}_j \frac{(\vec{x}_{j+1} - \vec{x}_j)}{\Delta\tau} - H(q_j, p_j) \right) \right]$$

$$= \left(\right) \exp \left(\frac{1}{\hbar} \left(\sum_j \vec{p}_j (\vec{x}_{j+1} - \vec{x}_j) - \Delta\tau H(q_j, p_j) \right) \right)$$

$$\Delta\tau = T/N \quad \checkmark$$

$$\frac{1}{\hbar} \sum p_j \Delta q - \Delta\tau \left(\frac{p_j^2}{2m} + \frac{1}{2} m \omega^2 q_j^2 \right)$$

$$\begin{aligned}
 & \int \mathcal{D}p_j e^{\left(\frac{1}{\hbar} q_{j+1} - q_j\right)^2 / 2 \left(\frac{\Delta t}{2m\hbar}\right)} \sqrt{\frac{\pi}{\Delta t/2m}} \\
 &= e^{\left(\frac{1}{\hbar} \sum_j (q_{j+1} - q_j)^2 \frac{m}{\Delta t}\right)} \sqrt{\frac{2m\pi}{\Delta t}} \\
 &= e^{-\frac{1}{\hbar} \Delta t \sum_j \frac{q_j^2 m}{2} \left(\sqrt{\frac{2m\pi}{\Delta t}}\right)} e^{-\frac{\Delta t}{\hbar} \sum_j \frac{1}{2} m \omega^2 q_j^2} \\
 &= e^{-\frac{1}{\hbar} \Delta t \sum_j \left(\frac{1}{2} m \dot{q}_j^2 + \frac{1}{2} m \omega^2 q_j^2\right)} \\
 &= \int d^3 x_i \int \mathcal{D}x(\tau) e
 \end{aligned}$$

$$\begin{aligned}
 &= \lim_{N \rightarrow \infty} \left(\right)^{1/2} \\
 &= \int d^3 x_i \int \mathcal{D}x(\tau) e^{-\frac{1}{\hbar} \int d\tau L_E \left(\frac{2m\pi}{\Delta t}\right)^{1/2} \frac{1}{(2\pi\hbar)^{3N}}}
 \end{aligned}$$

$$C = \sqrt{\frac{2m\pi}{\Delta t}} \frac{1}{(2\pi\hbar)^{3N}} \quad \Delta t = \hbar\beta = i\Delta\tau = i\frac{\tau}{N}$$

$$\Delta t = i\hbar\beta$$

$$C = \left(\frac{N2m\pi}{i\hbar\beta}\right)^{1/2} \frac{1}{(2\pi\hbar)^{3N}}$$

$$* V(x) = \frac{1}{2} m \omega^2 x^2 \quad \left| \quad x_j = \frac{1}{\sqrt{N}} \sum_{k=-M}^M y_k e^{i2\pi k j/N} \right.$$

$$M = (N-1)/2 \quad (y_0)$$

$$\text{D.O.F. : } 1 + 2 \times M = N \quad (y_k \neq 0)$$

$$y_i \in \mathbb{C}$$

$$\begin{aligned}
 x_{j+N} &= \frac{1}{\sqrt{N}} \sum_{k=-M}^M y_k e^{i2\pi(j+N)k/N} \\
 &= \frac{1}{\sqrt{N}} \sum_k y_k e^{2\pi i k} e^{i2\pi j k/N} = x_j
 \end{aligned}$$

- $k \in \mathbb{N}$ so $e^{2\pi i k} = 1$

$$X_{0+N} = X_0 \Leftrightarrow \boxed{X_N = X_0} \text{ as desired.}$$

$$X_J^* = X_J \Leftrightarrow \sum_{k=-N}^N y_k^* e^{-2\pi i k J/N} = \sum_{k=-N}^N y_k e^{2\pi i k J/N}$$

$$l \equiv -k \quad \sum_{l=-N}^N y_{-l} e^{-2\pi i l J/N} = \sum_{l=-N}^N y_l^* e^{2\pi i l J/N}$$

$$y_{-l} = y_l^*$$